FINAL-YEAR COMPUTER ENGINEERING · SIIT, THAMMASAT UNIVERSITY

SJ

Singha Junchan

สิงหา จุลจันทร์ / Pathum Thani, Thailand

I build security systems — and research how to make them provable.

Final-year Computer Engineering student at SIIT, Thammasat University (GPA 3.67/4.00, graduating May 2027). I lead the two-person team building TU Alert for Thammasat University (launching soon), and research security for cloud and AI — U-HAP won the Best Paper Award at IEEE JCSSE 2026 (second author) and FEDAC is under review at IEEE TDSC (first author). 1st in Thailand at the 2022 Olympiad in Informatics.

Scan or click for the full portfolio, code links and contact.

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Open to: internships in cybersecurity, cloud security or AI security from January 2027, in Thailand or abroad
full-time roles from May 2027

COMPETITIVE PROGRAMMING
#1
in Thailand: TOI18 national champion, 2022, after 1,000+ hours of training

CLIENT PROJECT
TU Alert
built for Thammasat University; I lead a team of two; security built in; launching soon

RESEARCH
2
2 accepted (U-HAP Best Paper at JCSSE 2026; ZKAudit at KST 2027) · 3 under review · 4 in progress

INDUSTRY
2
internships: Allianz Ayudhya (security, mentored by the CISO), Amity Solutions (AI research)

LEADERSHIP AND TEACHING
11
events organised; led SIIT First Meet 2024 (about 500 participants, 100 staff); founded two coaching platforms

CERTIFICATION
ISC2 CC
ISC2 Certified in Cybersecurity, 2025

01 What I bring

Security engineering, end to end

I design security mechanisms and build them: a zero-trust framework with a zero-knowledge verifier (first author) and a unified Kubernetes authorization webhook (second author).

Cloud-native by default

I test ideas on cloud deployments, not only on paper: Kubernetes clusters on AWS EKS, prototypes on EC2, containers with Docker.

AI agents, and how to limit them

I build AI agents that do useful work, and I research controls that keep them inside what a human allowed, even when they are attacked.

Leads, teaches and explains

I start things, run teams and explain hard ideas simply, to students, teammates and security teams.

02 Experience

Team Lead and Developer, TU Alert • Launching soon
Thammasat University (client project) / Rangsit campus, Pathum Thani — 2026

- Thammasat University hired a two-person student team to build TU Alert, a LINE-based campus incident-reporting system. I lead the team, and we both write the code.
- TypeScript monorepo (Hono, Bun, Prisma, MariaDB, Redis, Next.js, LINE LIFF); four roles, second-step staff sign-in, data retention and erasure, Semgrep and gitleaks in CI. Built to production standards; launching soon.

Information Security Intern **Allianz Ayudhya** / Bangkok, Thailand — Jun to Jul 2026

- Mentored by the Chief Information Security Officer (CISO) of Allianz Ayudhya, and worked with the Information Security team during university summer training.
- I built two AI-agent prototypes for third-party risk management. The supplier-side agent answers due-diligence questionnaires; its answers come with evidence, reasoning and confidence scores, and it flags gaps. The buyer-side agent guides business owners through the assessment.

Founder, Curriculum Author and Lead Instructor **TOI Tutor and ABIG Center (online)** / toitutor.in.th — 2022 to present

- Founded and lead two online learning platforms that coach students for Thailand's national programming olympiads.
- Wrote and taught COM300, a full C++ and algorithms course: 9 chapters, about 270 pages of notes and 128 recorded video lessons (Dec 2024 to Jan 2025).

R&D Intern, AI research team **Amity Solutions** / Bangkok, Thailand — Jul to Aug 2023

- Worked on the AI research and development team: researched and developed chatbot, data-analysis and prompt-engineering solutions, and studied how NLP and LLMs fit the full chatbot pipeline.

03 Research

Research supervisor: Prof. Somchart Fugkeaw, School of ICT, SIIT.

One question runs through most of my research: can a security check be cheap enough to run on every request, and still be provable afterwards? Each card shows the gap, what is new, what I built, where it helps, and its limits. Unpublished work shows the idea only.

RESEARCH MAP **Zero-trust access control** · **Zero-knowledge audit** · **Post-quantum security** · **AI-agent security** · **Security operations** · **Resilient cloud and edge**

FEDAC

● Under review · IEEE TDSC | First author of 4 | github.com/singhaju/FEDAC

Check the caller carefully once, then check a short-lived pass cheaply on every call

The gap In federated Kubernetes (several clusters, several organisations), the built-in RBAC is static: a role stays the same whichever service is called and however risky the request is. Adding risk-based sign-in, private cross-domain identities and tamper-evident policies usually puts heavy cryptographic or on-chain work on every request. The published designs we studied did not combine all of these without slowing requests.

What is new Split the expensive work from the per-request check. A hub verifies the caller once, choosing the sign-in strength per service by its sensitivity: single sign-on, a one-time code, a context risk check, then a zero-knowledge proof. It checks the policy, stored on IPFS, against a Merkle root held on a blockchain, then issues a short-lived token scoped to one service. A sidecar checks that token locally through Envoy in Istio, with no network call. If a policy is tampered with or a security event occurs, only tokens for the affected services are revoked.

What I built As first author in a team of four (two fellow students and Prof. Fugkeaw), I worked on the research design and built the implementation in Go and Rust: the hub, the Envoy external-authorization sidecar, four sign-in tiers including a Groth16 zero-knowledge verifier, targeted revocation and blockchain anchoring. I also re-implemented published designs and compared them with FEDAC on AWS EKS.

Where it helps Example: a compliance analyst at a financial firm opens an audit dashboard, a risk-model service and a trading service in one session. Each service asks for the sign-in strength it needs, each call is checked on the spot, and every policy change can be audited later.

Limits and next The first sign-in is still the slow step, so FEDAC suits services that call each other again and again. It was tested in one AWS region on a test blockchain, and the prototype signs private attributes but does not encrypt them yet. Hardware-backed keys and federation across regions are next.

RESULT **1.4 ms** median authorization latency on AWS EKS, about 7,100 requests per second (paper under review)

U-HAP

● Presented · IEEE JCSSE 2026 · Best Paper Award | Second author of 4 · presented the paper

github.com/singhaju/U-HAP

One place to write every Kubernetes access rule, with the same answer every time

The gap Kubernetes splits authorization across separate modules (RBAC, Node, webhook, ABAC) that do not know about each other, and RBAC can only grant access, never deny it. A rule that mixes models, such as "ops may delete pods in production only in office hours, unless the pod is critical", has to be stitched together, with no one place to see which rule wins. Rules get duplicated, and decisions depend on the order of checks.

What is new Treat authorization as a compiler problem. One graph model covers RBAC, ABAC, ACL and deny rules. Policies are compiled ahead of time into indexes per resource and action, so a request only looks at rules that can apply to it. Deny always wins, so every decision is deterministic.

What I built As second author in a team of four, I co-developed the Kubernetes authorization webhook and its policy model in Python, and checked that its decisions match Open Policy Agent on the same test requests. I presented the paper at JCSSE 2026 in Bangkok.

Where it helps Platform and security teams get one place to write and review every kind of access rule, and the same answer every time.

Limits and next Rules are scoped to one namespace today. Next: cluster-wide and cross-namespace rules, recompiling only what changed, and multi-cluster deployments.

RESULT **~2.7× faster** policy updates reach a first decision vs OPA at 2,000 policies, with identical decisions

MORE RESEARCH **ZKAudit** *Accepted · KST 2027.* Prove you may audit, without revealing your credentials. **MoCA: AI containment** *In preparation.* Keep a hijacked AI agent inside what its human allowed: an MCP gateway and resource-side checks. **ZK-ALBA** *In preparation · journal paper.* Zero-knowledge audit proofs bound to what each auditor may see. **Threshold-gated encrypted SIEM** *In preparation.* A SIEM that works on encrypted logs shared by companies that do not trust each other.

ALSO IN THE PIPELINE Three more first-author projects with Prof. Somchart Fugkeaw, two of them submitted: post-quantum authorization (ML-DSA and ML-KEM in a Go prototype), freshness proofs across clouds, and self-healing fog nodes. Details on the website.

04 Education and leadership

Education

B.Eng. in Computer Engineering — Aug 2023 to May 2027 (expected)
Minor in Cloud Computing and Cybersecurity. Sirindhorn International Institute of Technology (SIIT), Thammasat University GPA 3.67 / 4.00, first-class honours expected.

Leadership

11 student events organised · ~500 participants at SIIT First Meet 2024 · 100 staff at SIIT First Meet 2024

President, Sub Student Committee 32, SIIT — 2024 to present
Leads a committee that represents the student body and organises academic and social events.

Lead organiser, SIIT First Meet 2024 — 2024
About 500 participants and 100 staff. Organiser of 10 more student events, 2023 to 2025.

Team lead, TU Alert — 2026
Leads the two-person team building TU Alert for Thammasat University; both write the code.

Founder and Lead Instructor, TOI Tutor and ABIG Center — 2022 to present
Founded both olympiad coaching platforms and still leads them (details under Experience).

Project manager and product owner, Novel Nest — Team project
Led a team of four to build a digital library and creative hub for readers and writers.

05 Awards and certifications

AWARDS

2022 — **National Champion: 1st place in Thailand and Gold Medal**
Thailand Olympiad in Informatics (TOI18)

2026 — **Best Paper Award**
IEEE JCSSE 2026, Bangkok, for U-HAP (second author of 4)

2025 — **1st place overall**
Image Recognition Programming Challenge, Osaka Institute of Technology and SIIT

2023 to 2027 — **Full scholarship**
4-Year Continuing Academic Outstanding Student Program, SIIT

2022 — **Bronze Medal and Popular Vote**
SCIUS Forum 2022, Automated Cat Feeder

CERTIFICATIONS

✓ **ISC2 Certified in Cybersecurity (CC)** — 2025

planned — **CompTIA Security+**
Exam planned Dec 2026

06 Hands-on security

NCSA CTF Thailand Tournament 2026 — Thailand Cyber Top Talent 2026, Aug 2026. Competed in a team on web, cryptography, digital forensics, OSINT and mobile challenges.

SOSECURE Road to Cybersecurity 2025 (GEN6) — Eight modules with labs: routing and firewalls, Linux hardening, Python security tools, Metasploit and web attacks, SOC log analysis in Splunk and Wazuh.

Ongoing practice — TryHackMe and Hack The Box. Nmap scanning in a Kali Linux home lab. DataCamp courses on Azure and AWS (2025).

07 Projects

TU Alert, Novel Nest, Arduino retro console, Automated Cat Feeder, Velour Mall.

08 Skills

Used in my research	Zero Trust · Kubernetes (RBAC, ABAC, webhooks) · AWS (EKS, EC2) · Docker · Envoy and Istio · OPA / Rego · IPFS · Hyperledger Fabric
AI agents and AI security	LLM agent pipelines · MCP gateway · Prompt-injection testing · Prompt engineering · NLP chatbots
Shipping and operations	GitHub Actions CI · Semgrep · gitleaks · Playwright · pm2 on Linux · Backups and restore
Cryptography	zk-SNARKs (Groth16) · Merkle commitments · ML-DSA · ML-KEM
Security tools (labs and CTFs)	Nmap · Wireshark · Burp Suite · Metasploit · OSINT · OWASP Top 10
Programming	Python · Go · Rust · C++ · C · Java · Bash · TypeScript · SQL
Also familiar with	Splunk and Wazuh (SOC labs) · Azure and AWS basics (DataCamp)
Languages	Thai (native) · English (fluent) · Chinese (basic)

Let's talk

Looking for an internship from January 2027 and full-time roles from May 2027: security engineering, cloud security or AI security.

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